

A student team building an autonomous boat from the ground up.



HOW WE MODEL IT

First-principles mixed-integer linear programming

We build the model from physics up. Each constraint cuts the space of possible boats, and the optimizer finds the fastest one left. Hull resistance comes from **Savitsky's planing-hull model**. As we build and test, we keep updating the model and adding constraints: a hybrid of simulation and empirical data.

ENGINEERS FROM ACROSS UW-MADISON

Each segment of the boat is covered, and members have worked at Tesla, Northrop Grumman, Xcel Energy and Milwaukee Tool.

Industrial

Optimization, composites and manufacturing

Mechanical

Structures, CAD and fabrication

Electrical

Power, battery and high voltage

Computer

Embedded, firmware and software

OUR APPROACH

Hull

Plane well and stay light.

- Hybrid planing composite hull
- Shape tuned by optimization
- Resistance checked in CFD first

Electric

Every watt-hour to speed.

- High-voltage battery storage
- Propulsion sized by the model
- Propulsion and cooling co-designed

Autonomous

Manage drift and controls.

- GPS, IMU and LIDAR sensing
- Firmware protects the battery
- Software plans the route

TIMELINE

Oct 25	Current hull design frozen
Nov 15	Preliminary design review
Dec 1	Mechanical and electrical design frozen
December	Build the hull and composite parts
Early January	First float and tow test
January	Manufacture and assemble everything

Jan-Feb	Build likely spare parts, in parallel
End of January	Zeroth nautical mile
Early February	Slow drives on the water
Mid February	Start increasing speed
Until competition	Test as much as possible

Sponsor the build

We are looking for funds, parts, advisors and people to build our BOM as we design.

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CONTACT

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